Sales Forecasting for Nova Mart Ltd

An End-to-End Machine Learning Approach to Predicting Retail Sales

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1. Executive Summary

Nova Mart Ltd, a mid-sized U.S. retail supplier of Furniture, Office Supplies, and Technology products, has historically relied on spreadsheet trend lines and intuition to forecast monthly sales. As the business grew across regions and product categories, this approach produced recurring inventory problems: overstocking ahead of slow months and understocking ahead of seasonal peaks. This project builds a data-driven forecasting system to replace guesswork with a validated machine learning model, deployed as a usable application for the planning team.

Working from three years of historical order-level transaction data, the project covered the full data science lifecycle: data cleaning, exploratory analysis, feature engineering, time series diagnostics, comparison of seven forecasting models, hyperparameter tuning, and deployment. The final model — a tuned XGBoost Regressor — forecasts monthly sales within approximately 13.65% average error (MAPE), a substantial improvement over the spreadsheet-based approach it replaces, and was selected for its consistency across cross-validation folds rather than its performance on a single test split alone.

2. Business Context

2.1 Company Background

Nova Mart Ltd was founded a little over a decade ago as a regional office-supply retailer and has since grown into a multi-category retail business, selling Furniture, Office Supplies, and Technology products to Consumer, Corporate, and Home Office customers across four U.S. regions: West, East, Central, and South.

2.2 The Problem

- Recurring overstocking ahead of slow months, tying up cash in warehouse space
- Recurring understocking ahead of peak months, resulting in missed sales and frustrated corporate clients
- Finance team could sense seasonal spikes intuitively but had no reliable numbers to plan around
- Goal: replace guesswork with a data-driven forecasting system to guide inventory and staffing decisions

3. The Dataset & Data Cleaning Approach

Nova Mart supplied order-level transaction records spanning January 2015 to December 2018, including Order Date, Ship Date, Sales amount, Category, Sub-Category, Region, Segment, Customer, and Product fields.

3.1 Cleaning Steps

- Corrected data types — parsed Order Date and Ship Date as proper datetime objects
- Checked for and handled missing values
- Removed duplicate order records
- Standardized categorical fields (e.g. consistent region and category naming)
- Aggregated transaction-level data into a monthly sales time series, the basis for all subsequent analysis

4. Exploratory Data Analysis

4.1 Trend & Seasonality

- Clear upward sales trend from 2015 to 2018, with a temporary dip in 2016
- Strong seasonality: a major peak in November/December (holiday season) and a secondary peak in March
- Tuesday and Saturday are the strongest sales days of the week; Thursday is the weakest

4.2 Category, Segment & Distribution

- Technology — particularly Phones — leads category revenue; Consumer leads by customer segment
- Individual sale amounts are right-skewed; large orders are legitimate bulk/corporate purchases, not data errors
- No strong linear correlation was found between sales and the raw numeric fields, motivating a dedicated feature engineering phase

5. Feature Engineering & Time Series Analysis

5.1 Engineered Features

- Calendar features: Month, Quarter, Year, and a cyclical sine/cosine encoding of Month
- Lag features: sales from 1, 2, 3, and 12 months prior
- Rolling window features: 3-month and 6-month rolling mean and standard deviation
- Change features: month-over-month percentage change and raw difference
- IsHoliday flag marking months containing a U.S. public holiday

5.2 Stationarity Testing & Decomposition

An Augmented Dickey-Fuller (ADF) test indicated the monthly series was stationary ($p = 0.0003$), while a KPSS test indicated it was non-stationary ($p = 0.02$). This apparent contradiction is a well-known signature of a trend-stationary series — one with a real, steady deterministic trend but no runaway, unit-root behavior. A formal seasonal decomposition confirmed this: a steady upward trend and a seasonal component that repeats reliably every 12 months, consistent with the November/December and March peaks identified during EDA. First-order differencing fully stabilized the series (ADF $p \approx 0.0000$), confirming that a single differencing pass ($d = 1$) is sufficient for any classical model requiring stationary input.

6. Modelling Approach & Results

Seven models were trained and compared: a Naive baseline (carries forward the prior month's sales), Linear Regression, Random Forest, XGBoost, Prophet, ARIMA, and SARIMA. Each was evaluated on Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), using both a single chronological train/test split and 4-fold time-series cross-validation, to confirm results were not the product of a single lucky split.

6.1 Cross-Validation Results (mean across folds)

Model	Mean MAE	Mean MAPE

XGBoost	16,063	27.30%
SARIMA	18,958	35.70%
Naive	21,773	43.16%
ARIMA	22,145	40.15%
Linear Regression	41,534 (high std)	93.52%
Prophet	82,096 (high std)	167.67%

Table 1. Multi-window cross-validation performance, ranked by mean MAE.

Linear Regression scored best on a single test split but was the worst and most volatile model across cross-validation folds — a classic sign of overfitting to one particular train/test boundary rather than genuinely generalizing. XGBoost was the most consistent performer across every fold, making it the more trustworthy choice for deployment despite not topping the single-split leaderboard.

6.2 Hyperparameter Tuning & Final Comparison

XGBoost and SARIMA — the two strongest cross-validation performers — were tuned via RandomizedSearchCV with time-series-aware cross-validation. Tuning improved XGBoost's test-set MAPE from 14.34% to 13.65%. SARIMA's tuned result was substantially worse than its untuned baseline, most likely because the hyperparameter search overfit to the limited number of folds available for a short monthly series.

Model	MAE	RMSE	MAPE
XGBoost (Tuned)	10,790	14,825	13.65%
XGBoost (Untuned)	11,210	15,127	14.34%
Random Forest	12,550	15,912	16.60%
SARIMA (Untuned)	15,126	20,098	20.18%
Naive	17,681	22,082	22.35%
ARIMA	24,641	33,409	28.20%
Prophet	23,096	27,068	35.77%
SARIMA (Tuned)	26,205	34,874	30.43%

Table 2. Final model comparison on the held-out test set, sorted by MAE.

6.3 Selected Model

The final model selected for deployment is the tuned XGBoost Regressor (n_estimators=500, max_depth=3, learning_rate=0.05, subsample=0.6, colsample_bytree=1.0, min_child_weight=1), achieving MAE of $10,790, RMSE of $14,825, and MAPE of 13.65% on the held-out test period. The dominant predictive features were the month-over-month sales difference and the same-month-last-year lag value — directly reflecting the trend and annual seasonality structure identified during EDA and time series decomposition.

7. Business Insights & Recommendations

- Months containing a U.S. public holiday average approximately 31% higher sales than non-holiday months — a concrete figure to build into the inventory calendar
- Recommend increasing inventory and staffing 1–2 months ahead of predicted seasonal peaks, particularly for Technology and Furniture — the highest-revenue categories and those most exposed to stockout risk during the holiday rush
- The model forecasts monthly sales within roughly 13.65% average error, a meaningful and measurable improvement over the prior spreadsheet-based estimates

8. Deployment

The final model was retrained on the full historical dataset and deployed as an interactive Streamlit application, giving Nova Mart's planning team direct access to monthly forecasts without needing to open a notebook. The app displays historical sales, generates forecasts for a user-selected number of months ahead, and returns each forecast alongside a 90% prediction interval and a plain-language interpretation.

Forecasts beyond the first month are generated recursively — each predicted month's value is fed back in to compute the following month's lag and rolling features. This means forecast error compounds with horizon length, so short-horizon forecasts (1–2 months) are considerably more reliable than longer ones; this caveat is surfaced directly in the app and documented in the project README.

8.1 Tech Stack & Repository

- Python, pandas, NumPy, scikit-learn, XGBoost, statsmodels, Prophet — analysis and modelling
- Streamlit — deployed forecasting application
- Repository includes: app.py (Streamlit app), model_utils.py (feature-generation logic used at inference time), the trained model artifact, feature column metadata, historical sales data for seeding forecasts, and a README covering setup and deployment instructions
- Deployed via Streamlit Community Cloud, connected directly to the project's GitHub repository

9. Limitations & Future Work

- The model was trained and validated on 2015–2018 data only; performance should be re-validated against more recent sales before any long-term production reliance
- MAPE is computed excluding months with zero actual sales to avoid divide-by-zero errors — a limitation of the metric on sparse data rather than of the model itself
- Recursive multi-month forecasting compounds error with horizon length; a direct multi-step forecasting approach (training separate models per horizon) could reduce this in future work
- SARIMA's seasonal order search used a weekly seasonal period (m=7) during an earlier daily-grain experiment in error; a corrected search using m=12 on monthly data is a worthwhile follow-up before ruling out classical seasonal models entirely
- Category-level and region-level forecasting (rather than one blended company-wide total) is a natural next step, since Technology and Furniture behave differently from Office Supplies seasonally
- Incorporating external signals — marketing calendar, promotions, macroeconomic indicators — could further improve accuracy beyond what internal sales history alone can capture

10. Conclusion

This project demonstrates a complete, production-oriented forecasting workflow: from raw transaction data, through rigorous exploratory and time series analysis, to a validated, tuned machine learning model deployed as a tool Nova Mart's planning team can use every month. The emphasis on cross-validation over single-split evaluation, and the transparent handling of a case where hyperparameter tuning made a model worse, reflect a deliberately careful and defensible modelling process rather than one optimized to produce the most impressive-looking single number.

Appendix A: Metric Definitions

The following metrics are used throughout this report to evaluate forecast accuracy. All three are computed on the held-out test period, in the original Sales unit (US dollars) or as a percentage.

MAE — Mean Absolute Error

The average absolute difference between predicted and actual sales, in dollars. An MAE of $10,790 means forecasts are, on average, about $10,790 away from the true monthly sales figure, regardless of whether the model over- or under-predicted.

RMSE — Root Mean Squared Error

Similar to MAE but squares errors before averaging, then takes the square root. This penalizes large errors more heavily than small ones, so RMSE is always ≥ MAE; a wide gap between the two indicates the model occasionally makes a few large errors rather than being consistently off by a moderate amount.

MAPE — Mean Absolute Percentage Error

Expresses the average error as a percentage of actual sales, making it easier to communicate to a non-technical audience (“forecasts are typically off by about 13.65%”) and to compare accuracy across periods of different sales volume. Months with zero actual sales are excluded from this calculation to avoid division by zero.

Appendix B: Models Evaluated

- Naive — a simple baseline that predicts next month's sales will equal the current month's sales; any useful model should beat this
- Linear Regression — a straight-line relationship between engineered features and sales
- Random Forest — an ensemble of decision trees that averages many independent predictions
- XGBoost — a gradient-boosted tree ensemble that builds trees sequentially, each correcting the previous trees' errors; the selected production model
- Prophet — an additive time series model developed for business forecasting, designed to handle trend and seasonality with minimal tuning
- ARIMA — a classical statistical model that forecasts from a series' own past values and past errors
- SARIMA — ARIMA extended with an explicit seasonal component, intended to capture repeating annual patterns